

Program: Diploma in Food Processing Technology	
Course Code: 3424	Course Title: Food Microbiology I
Semester: 3	Credits: 3
Course Category: Program core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

1. To understand the history and scope of microorganisms in food
2. To understand the isolation, growth, quantification, and culturing of microorganisms
3. To impart knowledge on food poisoning and food borne pathogens

Course Prerequisites:

Topic	Program/CourseName
Basic knowledge of micro-organism	Micro organism in food materials

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Recognize the history of food microbiology, basics of Microscopy, major microorganisms in food	14	Understanding
CO2	Understand the Microorganism isolation, growth, quantification, and culture.	16	Understanding
CO3	Identify the factors and microbes involved in food spoilage; food borne pathogens, food poisoning and microbial toxins.	14	Understanding
CO4	Summarize the Microbial quality assurance systems in food industry	16	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Recognize the history of food microbiology, basics of Microscopy, major microorganisms in food		
M1.01	Discuss the history of microbiology of Food, scope of food microbiology	5	Understanding
M1.02	Outline the basics of Microscopy and its different types	5	Understanding
M1.03	Summarize microorganisms associated with food	4	Understanding
Contents: History of microbiology of Food; Food microbiology and its Scope. Basics of Microscope; different types of Microscopy (Bright –Field Microscopy, Dark-Field Microscopy), Types of microorganisms associated with food: bacteria, fungi, & virus			
CO2	Understand the Microorganism isolation, growth, quantification, and culture		
M2.01	Explain Microorganism isolation and its methods	5	Understanding
M2.02	Understand Growth curve and its phases	5	Understanding
M2.03	Discuss the microorganism quantification and culture	5	Understanding
	Series Test –I	1	
Contents: Methods of Isolation, preservation and maintenance of pure cultures, Growth curve,			
CO3	Identify the factors and microbes involved in food spoilage; food borne pathogens, food poisoning and microbial toxins.		

M3.01	Identify Factors affecting food spoilage	4	Understanding
M3.02	Enumerate Food poisonings, food borne illness and toxins	5	Understanding
M3.03	Outline the major food groups and microorganisms involved	5	Understanding
Contents: Factors affecting spoilage of foods, Micro flora associated with various food groups, . Food poisonings - due to pathogens, important features, Microbial toxins, Bacterial agents of food borne illness			
CO4	Summarize the Microbial quality assurance systems in food industry		
M4.01	Summarize the microbiological quality assurance systems	5	Understanding
M4.02	Recognize GMP and HACCP	5	Understanding
M4.03	Review Various standards in microbiological criteria	5	Understanding
	Series Test–II	1	
Contents: Microbial quality assurance systems in food industry, Codes of Good Manufacturing Practices, HACCP food standards, Quality control using microbiological criteria			

Text / Reference

T/R	Book Title/Author
R1	James M.J. (2000) Modern Food Microbiology, 5th Edition, CBS Publishers
R2	Barnart, G. J. (1997) Basic Food Microbiology, CBS Publishers
R3	William Carroll Frazier and Dennis C Westoff (2017) Food Microbiology, 5th Edition
R4	Bibek Ray (1996) Fundamental Food Microbiology, CRC Press
R5	M. R. Adams, M. O. Moss. 2008 Food Microbiology, 2nd Edition

Online Resources

Sl.No	WebsiteLink
1	https://faculty.uobasrah.edu.iq/uploads/teaching/1648242688.pdf
2	https://www.britannica.com/science/growth-curve
3	https://www.cdc.gov/foodsafety/foodborne-germs.html
4	https://ispe.org/initiatives/regulatory-resources/gmp/what-is-gmp